

INDEPENDENT SCIENTIFIC REVIEW

Coherence Collapse Theory (CCT) v2.0

*An informational phase-transition framework of psychosis:
analysis, reconstruction, and evidential status*

Source specification: Landry, M. (2026). CCT v2.0, revised following formal audit.

Document type: scientific overview and critical analysis. Not an endorsement, protocol, or validation.

Classification used here: mid-level computational hypothesis set, consistent with the source's own scope statement.

1. Purpose and standard of assessment

This document reconstructs Coherence Collapse Theory version 2.0, states what the specification actually claims, and assesses those claims against the standards appropriate to a computational model in psychiatry: identifiability of parameters, dimensional consistency, existence of estimators, separation of analogy from measurement, and explicit falsification conditions. The source text is treated as a research specification, not as established science. That reading is required by the specification itself, which describes CCT as a mid-level computational model that “generates hypotheses” and “does not yet have the status of a confirmed theory.”

The review does not dispute the phenomenological observation that acute psychosis often involves altered self–world boundaries, heightened salience of prediction error, and disturbed temporal organization. Those phenomena are already the subject of a substantial computational-psychiatry literature. The question is whether CCT v2.0 adds a coherent, measurable, and non-redundant formalization.

2. Executive assessment

Version 2.0 is a material improvement on the audited predecessor. Sign errors in the boundary index, the unbounded continuum-coherence construct, the non-falsifiable conservation axiom, the seven-dimensional state vector, the undefined master functional, and the unrestricted multiscale isomorphism are either corrected or retired. The revision also states, correctly, that numerical thresholds require calibration and that intervention mappings are hypotheses rather than protocols.

The remaining document is best read as a structured hypothesis linking four quantities—precision gain, boundary permeability, temporal integration, and noise sensitivity—to psychotic phenomenology, together with a proposed measurement stack drawn from existing methods (hierarchical Gaussian filtering, conditional-independence discovery, autocorrelation, psychophysics, and early-warning statistics). It is not a derivation from first principles, not a demonstrated phase transition, and not a diagnostic instrument.

Three structural problems survive the revision and limit scientific grade. First, the corrected bookkeeping equation adds a dimensionless insulation index to excess entropy, which is not dimensionless in the same units; this repeats the dimensional inhomogeneity the audit was meant to remove. Second, several “healthy ranges” and deviation cut-offs are still stated as numbers, in tension with the claim that thresholds are not known prior to calibration. Third, the named estimators do not, as specified, identify the named parameters: PCMCi detects conditional dependence, it does not estimate normalized conditional mutual information; the hierarchical Gaussian filter parameter ω is not identical to a free precision-gain coordinate π . Until those identification maps are written, the psychotic state vector is not yet an estimand.

3. What the specification claims, and what it withdraws

3.1 Bounded claims

CCT v2.0 advances three claims. The computational claim is that psychotic phenomenology is associated with a characteristic pattern of deviations in precision weighting, boundary integrity, and temporal integration inside predictive-processing architectures. The dynamical claim is that the prodrome-to-acute transition involves a change in the stability of the self-model attractor landscape, detectable by early-warning signals. The measurement claim is that the deviations can be estimated from neurophysiological and behavioral data, and that their trajectories predict clinical outcomes.

3.2 Explicit withdrawals

The specification withdraws the claim that psychosis is a universal phase transition identical across scales; any derivation from first principles; prior knowledge of numerical thresholds; application to quantum or cosmological systems without a shared estimator; and the idea that parameters can be directly set by intervention. Cross-domain rows for quantum and cosmological systems are retired. The master functional, spectral magic numbers (period 4–7, Lyapunov exponent +0.27, half-life 1.1 iterations), invented coordinate examples, and efficacy claims for a “coherence restoration cascade” are retired or gated on future trials. These withdrawals are scientifically appropriate.

4. Formal reconstruction

4.1 Boundary coherence

The version 1.0 index equalled the uncertainty coefficient $I(\text{internal}; \text{external}) / H(\text{internal})$. That quantity is large when internal states are predictable from external states, which is the opposite of insulation. Version 2.0 conditions on blanket states:

$$CI_B = 1 - I(s_{int}; s_{ext} | s_{blanket}) / H(s_{int} | s_{blanket})$$

For discrete variables, conditional mutual information is at most the relevant conditional entropy, so the ratio lies in $[0, 1]$ and $CI_B = 1$ denotes conditional independence of internal and external states given the blanket. That is the correct information-theoretic reading of insulation in a Markov-blanket partition (Pearl; Friston’s use of blankets is related but not identical to this estimator). The correction of the sign error is therefore accepted.

Two caveats remain. Differential entropy and differential mutual information do not obey the same bounds; the specification correctly requires quantization or a stated reference measure for continuous series, but does not specify the partition, the binning, or the bias correction. More importantly, the proposed estimator—PCMCI (Runge et al., 2019)—is a constraint-based causal-discovery procedure for conditional independence in time series. It can reject or fail to reject independence. It does not return the normalized mutual-information ratio in Eq. 2.1. A magnitude estimator (for example a k-nearest-neighbour conditional mutual information estimator, with a preregistered quantization) must be specified separately, and the blanket states must be designated before estimation, not discovered post hoc from the same series.

4.2 Excess entropy in place of continuum coherence

The former continuum-coherence index was a normalized mutual information, hence bounded, while the prose claimed divergence to infinity. Version 2.0 replaces it with excess entropy, or predictive information:

$$E = I(X_{past}; X_{future})$$

Excess entropy is a standard complexity measure (Grassberger, 1986; Crutchfield and Feldman). It is nonnegative for discrete processes and typically grows with the amount of structured temporal memory. The revised hypothesis—that psychosis involves reduced excess entropy (temporal disorganization) despite elevated local correlation—is coherent and distinct from the retired claim. Block-entropy and bias-corrected transfer-entropy estimators are appropriate in principle, but finite-sample bias is severe for neural and symptom series. The analysis window, embedding, and surrogate-data null must be preregistered. Local correlation and excess entropy are not identified by the same statistic; both need operational definitions if the “elevated local, reduced global” dissociation is to be tested.

4.3 Continuity equation: residual problem

The version 1.0 conservation axiom was non-falsifiable because psychosis was defined as its violation. Version 2.0 replaces it with an approximate continuity statement whose residual is the object of study:

$$d/dt (CI_B + E) = \sigma_{residual}$$

Reframing an axiom as a bookkeeping residual is the right epistemic move. The equation as written is not yet usable. CI_B is dimensionless on [0, 1]. Excess entropy is an information quantity whose magnitude depends on alphabet size, sampling rate, and the past–future split. Their sum is dimensionally inhomogeneous—the same class of error cited against the retired master functional. A scientifically usable residual would be defined on dimensionless, separately normalized series, or as a vector residual, not as a scalar sum. The proposed falsifier (acute-psychosis residual above the 95th percentile of healthy controls under equivalent task demand) is testable only after the residual is defined, the task is matched, and medication, sleep, and movement confounds are controlled.

5. Psychotic state vector

The state vector is reduced from seven incompatible coordinates to four parameters with claimed estimators:

$$PSV = (\pi, \beta, \tau, \nu)$$

Boundary permeability is defined as $\beta = 1 - CI_B$, which inherits the identification problems of CI_B but is at least dimensionally consistent with it. The other three coordinates require an explicit measurement model before the vector can be estimated.

Symbol	Intended quantity	Specification gap
π	Precision gain: weight on prediction errors relative to priors. Cited estimator: hierarchical Gaussian filter (Mathys et al.). Deviation stated as $\pi > 3.0$; healthy range [0.5, 2.0].	HGF does not contain a free parameter named π with that range. Perceptual uncertainty is carried by parameters such as ω (tonic volatility of the second level) and by trial-wise precision ratios. These are model- and task-dependent. A mapping $\pi = g(\omega, \kappa, \text{task})$ must be written, or the symbol should be retired in favour of the HGF parameter actually estimated.
β	Boundary permeability, $1 - CI_B$. Estimator: PCMCI on multimodal series. Deviation stated as $\beta > 0.7$.	PCMCI outputs conditional-independence decisions and, in some implementations, partial-correlation weights. It does not output CI_B . Blanket, internal, and external sets are undefined. The cut-off 0.7 is an uncalibrated constant.
τ	Temporal integration, operationalized as autocorrelation half-life, healthy range [0.3, 1.0] s, psychotic deviation $\tau < 0.1$ s. Alternative: delayed-embedding dimension.	Half-life and embedding dimension are different estimands. A half-life of 0.1 s is plausible for some neural signals and implausible for symptom series sampled daily. The signal, sampling rate, and preprocessing (including filtering, which induces autocorrelation) are unspecified.
ν	Noise sensitivity, inverse signal-to-noise threshold, healthy [0.02, 0.05], deviation > 0.08 . Estimator: psychophysical discrimination threshold.	A threshold from a staircase procedure is a task-specific Weber-like quantity, not a universal inverse signal-to-noise ratio. Modality (auditory, visual, proprioceptive) must be fixed. Elevated thresholds also occur in non-psychotic sensory impairment.

Table 1. Identification gaps in the revised psychotic state vector. Ranges are those stated in the source specification; they are not empirical estimates.

The reduction from seven coordinates to four is a genuine improvement, and retiring spectral radius, noise fraction, and regularization as primary coordinates avoids symbol collision. The vector is not yet a coordinate system: there is no stated inner product, no statement of conditional independence among coordinates, and no measurement-error model. For a logistic risk model these omissions matter, because collinear estimators will inflate coefficient variance and make external validation unstable.

6. Risk model, early-warning signals, and the cascade

6.1 From phase transition to logistic risk

Version 2.0 abandons a discrete AND/OR threshold and writes the probability of an episode as a logistic function of a weighted sum of the four parameters, with weights and bias to be estimated on a training cohort and validated externally. That is an ordinary supervised-risk model. It is estimable. It is not, by itself, evidence of a phase transition. A logistic function imposes linear log-odds and does not represent bistability, hysteresis, or critical slowing. If the dynamical claim is retained, the risk model and the dynamical model need to be related: for example, by showing that the logistic weights correspond to a section through a fitted potential, or by comparing the logistic model with a cusp or other bifurcation model on the same longitudinal data. The specification notes hysteresis as a prediction (relapse threshold below remission threshold for at least one parameter) and gives a falsifier. That prediction is separable from the logistic equation and should be tested separately.

6.2 Early-warning signals

The dynamical predictions follow Scheffer and colleagues (2009): rising lag-1 autocorrelation, rising variance, and shifts in skewness and kurtosis ahead of a transition. The proposed lead times—roughly one to four weeks—are labelled as hypotheses, which is correct. Application to psychosis is not straightforward. Symptom series in clinical high-risk designs are often weekly or sparser; autocorrelation estimates at that sampling rate have wide confidence intervals. Treatment changes, sleep disruption, and substance use produce variance rises that mimic critical slowing. A credible test requires a preregistered indicator, a null that includes non-critical nonstationarity, and a comparison against existing clinical high-risk instruments rather than against chance alone.

6.3 Cascade hypothesis

The ordered cascade π then β then τ is restated as a temporal-precedence hypothesis, to be tested by cross-lagged panel models or Granger causality, and falsified if boundary-permeability changes precede precision-gain changes in a majority of prodromal subjects. This is a well-formed hypothesis. Granger causality on sparsely sampled clinical series is a weak test of mechanism; a stronger design is a task-based hierarchical model in which volatility, sensory precision, and temporal binding are estimated in the same session and ordered within subject. The falsifier should also state the estimand and the handling of missing visits, otherwise “majority of subjects” is undefined.

7. Phenomenology, hallucinations, and the index

Six phenomenological signatures are mapped to computational substrates. Four of the mappings are at least directionally testable: elevated precision weighting, elevated boundary permeability, reduced temporal autocorrelation, and unstable identity measures or rivalry switching. “Multiversal simultaneity” is retained as description and retired as a formal construct. That distinction should be maintained in any figure or abstract; descriptive language must not re-enter the estimand list.

Auditory verbal hallucinations are remodelled as a Dirichlet-process mixture over latent policies, with component number inferred rather than fixed, and with model comparison against corollary-discharge and inner-speech accounts. This is an improvement on a fixed “three voices” claim. It is not yet a measurement model of hallucinations. A Dirichlet-process prior encodes a modelling preference for an unknown number of components; it does not establish that voices are policies. Independent-component or mixture counts in magnetoencephalography during hallucinations are not counts of voices. The falsifier—if a unified corollary-discharge model fits better, the fragmentation hypothesis fails for that dataset—is acceptable provided the model class, the data partition, and the score (WAIC or ELPD, as the generative-model section suggests) are fixed in advance.

The psychotic state index is a logistic function of the four parameters, interpreted as a probability, with calibration by isotonic regression or Platt scaling and utility judged by decision curves. That is the correct interpretation of a risk score. Version 1.0’s dependence on a single noise term is removed. Clinical prediction C1 sets a bar of AUC greater than 0.70 for episode within three months. Discrimination at that level is modest and can be achieved by existing symptom scores; a useful claim requires incremental value over a clinical high-risk baseline, reported as change in AUC or, preferably, as net

benefit, with an external cohort.

8. Intervention mapping and cross-domain analogies

The coherence-restoration sequence is explicitly demoted from protocol to research program. Target classes are paired with mechanism hypotheses: D2 antagonists with reduced precision weighting, grounding techniques with rebuilt blanket statistics, narrative therapy with restored autocorrelation, and sensory modulation with reduced noise sensitivity. The specification states that standard of care takes precedence and that efficacy claims await randomized trials. Those constraints are necessary and should be treated as binding.

The dopamine–precision hypothesis has a published computational-psychiatry background (Adams, Stephan, Friston, and related work on aberrant salience by Kapur, and on Bayesian accounts of positive symptoms by Fletcher and Frith). CCT does not add evidence for that hypothesis. A pre/post HGF comparison on a specified task would be a test, not a confirmation. Grounding, narrative therapy, and sensory modulation do not currently have identification theorems linking them to CI_B , τ , or v . Listing them in a table does not create a mechanism.

Cross-domain claims are restricted to shared dimensionless estimators computed with the same code against a preregistered threshold. Quantum and cosmological rows are removed. The artificial-intelligence analogy is narrowed to reward-model disagreement, policy divergence from a reference, and uncertainty miscalibration, with a falsifier if reward hacking occurs without elevated reward-model disagreement in a preregistered agent study. This is a legitimate analogy only if the word “psychosis” is not used as a label for agent failure. The quantities named are standard monitoring statistics in reinforcement learning; they do not share an estimator with the clinical state vector as currently defined. The social-network row remains programmatic.

9. Philosophical coda

The ethical correction is sound: a rule that boundary coherence must be nondecreasing forbids sleep, play, grief, and learning, and is replaced by a non-maleficence constraint against driving CI_B below a healthy range. The range is not yet known, so the constraint is not yet operational. The optimization slogan equating selfhood with constrained conditional-entropy minimization is heuristic. The domain of S , the environment measure, and the feasibility of the constraints are unspecified. It should not be cited as an equation of the theory.

The restriction on “psychosis as revelation” is also sound. Elevated salience is a computational state, not calibrated knowledge. Any formal account that omitted distress, functional impairment, and duration would fail to distinguish disorder from unusual experience. The specification records that constraint; analyses and figures should not drop it.

10. Relation to the existing literature

The cited empirical core is real and is the appropriate foundation. Kapur (2003) framed psychosis as aberrant salience. Fletcher and Frith (2009) gave a Bayesian account of positive symptoms. Adams, Stephan, Brown, Frith, and Friston (2013) set out a computational anatomy in which failures of precision at different levels of a hierarchy produce different symptom classes. Mathys and colleagues provided the hierarchical Gaussian filter used to estimate learning under uncertainty. Stephan, Baldeweg, and Friston linked synaptic plasticity and dysconnection to schizophrenia. Scheffer and colleagues formalized early-warning signals for critical transitions in other systems. Runge and colleagues provided PCMCi for time-series causal discovery. None of these sources establishes a four-parameter psychotic state vector, a boundary-coherence index of the form in Eq. 2.1, or a continuity residual whose elevation defines psychotic transition.

CCT v2.0 is therefore a synthesis and a proposed reparameterization, not a summary of results. Its incremental contribution, if any, will be the joint estimation of precision, blanket insulation, temporal memory, and noise sensitivity in the same individuals, with preregistered precedence and outcome tests. Joint estimation is not provided in the specification.

11. Audit response: what was fixed

Audited defect	v2.0 response	Status in this review
Sign error in boundary index	Conditioned on blanket states	Correction accepted; estimator not identified
Continuum coherence claimed to diverge	Replaced by excess entropy	Correction accepted
Conservation axiom non-falsifiable	Residual bookkeeping equation	Epistemic fix accepted; sum still inhomogeneous
Seven incompatible dimensions	Reduced to four	Reduction accepted; identification incomplete
Impossible and arbitrary coordinates	Retired or replaced	Accepted; new cut-offs remain uncalibrated
Spectral magic numbers	Retired	Accepted
Undefined master functional	Replaced by a generic state-space model	Accepted as a placeholder, not a model
Multiscale isomorphism	Restricted to shared estimators; quantum and cosmology retired	Accepted
Index dominated by one term	Logistic function of all four parameters	Accepted as a form; unestimated
Efficacy claims inside the theory	Gated on trials	Accepted
Monotone boundary rule	Non-maleficence constraint	Accepted; not yet operational
Self-citation only	Primary literature added	Citation set is appropriate but not exhaustive

Table 2. Disposition of the principal audit items.

12. What remains speculative

The specification's own appendix is accurate and should be read as part of the abstract, not as a footnote. The cascade order is untested. The agent analogy requires preregistered studies and a shared estimator it does not yet have. Social-coherence moderation requires network data. Multiscale isomorphism is programmatic. No cohort has been used to calibrate PSV coordinates, weights, or decision thresholds. In addition to those admissions, the present review finds that Eq. 2.3 is dimensionally inhomogeneous, that PCMCi and the hierarchical Gaussian filter are not yet measurement models for β and π , and that stated numerical ranges contradict the calibration gate.

13. Minimum scientific program

A version that could be preregistered would need the following, in order. None of these steps is a clinical recommendation.

1. Fix the observation streams: one behavioral task with a published HGF implementation, one neurophysiological series, and one symptom series, each with sampling rate and exclusion rules.
2. Replace π with the HGF parameter actually estimated, or publish the function that maps HGF output onto π , including units.
3. Define the Markov partition before seeing the data. Estimate conditional mutual information with a stated estimator and bias correction. Do not equate a PCMCi rejection decision with CI_B .
4. Choose one temporal estimand. If it is an autocorrelation half-life, state the series and the filter settings. Do not also treat embedding dimension as the same parameter.
5. Drop the scalar sum in Eq. 2.3, or normalize both terms to a common dimensionless scale and show invariance to that normalization.
6. Remove all numerical healthy ranges and cut-offs from the confirmatory text. Estimate them on a training cohort and freeze them before external validation.
7. Specify the clinical baseline model. Report incremental discrimination and net benefit, not AUC against chance.

8. Separate the dynamical falsifiers (critical slowing, hysteresis, cascade order) from the logistic risk model, and power each comparison.
9. Keep intervention language in a hypothesis register. Do not attach efficacy, sequence, or dosing claims to the theory.

14. Conclusion

CCT v2.0 is a corrected specification for a research program. It successfully retires the non-falsifiable and dimensionally broken elements of version 1.0 that were identified by audit, and it places its remaining claims under estimators and falsifiers drawn from computational psychiatry and dynamical systems. That is the right direction of revision.

It does not yet constitute a scientific theory of psychosis. The central state vector is not identified by the methods named for it. The continuity equation remains dimensionally inhomogeneous. Numerical thresholds re-enter the text after being declared unknown. The phase-transition language is stronger than the logistic risk model that replaced it. Phenomenological mappings, intervention tables, and agent analogies are hypotheses pending shared estimators and preregistered tests.

The appropriate citation status is: Landry (2026), CCT v2.0, a mid-level computational hypothesis set, uncalibrated, not a diagnostic system and not a treatment protocol. Further use should proceed by measurement-model specification and external validation, not by elaboration of the formalism.

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Scope note. This review analyses a theoretical specification. It does not diagnose, recommend treatment, or assign clinical meaning to any individual experience. Standard psychiatric care is independent of this framework.